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VULNERABILITY MANAGEMENT AND APPLICATION SECURITY CAPSTONE PROJECT ASSIGNMENT

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TECHBRIDGE CONSULTING LIMITED

Vulnerability Assessment, Risk-Based Prioritization & Remediation Management

Engagement Element	Details
Student Role	Vulnerability Management Analyst / Security Consultant
Engagement Type	Pre-Production Vulnerability Assessment
Client	TechBridge Consulting Limited
Industry	Financial Technology
Primary Objective	Identify, validate, prioritise and communicate vulnerability exposure before a planned production release.

Professional Context

This assignment simulates a client-facing vulnerability management engagement. You are expected to exercise professional judgement, validate evidence and communicate risk clearly. The objective is not to produce the largest number of scanner alerts.

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1. Engagement Background

TechBridge Consulting Limited is a UK-based financial technology company headquartered in Manchester. The company provides digital payment services, short-term consumer lending, virtual accounts, merchant payment services and mobile financial services to individuals and small businesses.

Founded six years ago, TechBridge Consulting Limited has experienced rapid growth following increased adoption of its digital services. The organisation currently employs approximately 215 people, serves 128,000 registered customers, supports about 46,000 monthly active mobile users and has approximately 3,800 registered merchants. Annual transaction value is approximately £310 million.

Approximately 82% of customer interactions now occur through TechBridge Consulting Limited's web and mobile platforms. The company is preparing for further digital expansion and expects transaction volumes to increase substantially over the next 12 months.ths.

2. Regulatory and Compliance Context

TechBridge Consulting Limited processes personal and financial information and operates within a regulated financial-services environment. Its security programme must therefore consider regulatory, legal and contractual obligations.

- UK General Data Protection Regulation (UK GDPR)
- Data Protection Act 2018
- Applicable Financial Conduct Authority (FCA) requirements
- FCA operational resilience expectations
- PCI DSS obligations where cardholder data is processed
- Contractual security obligations with banking and payment partners

Legal and Compliance have advised senior management that vulnerabilities affecting customer information, authentication services or payment-related systems could result in regulatory scrutiny, operational disruption, financial loss and reputational damage.

3. Business and Technology Environment

3.1 Customer Web Services

TechBridge Consulting Limited's customer-facing web services support activities including:

- Customer registration and profile management
- Loan application and account servicing
- Transaction-history access
- Selected customer transactions
- Payment-information management
- Supporting web and application services

3.2 Mobile Services

Mobile services are strategically important to TechBridge Consulting Limited. Android represents approximately 71% of the company's mobile-user base. The organisation is reviewing new and experimental Android builds before deciding whether they should progress toward production..

Mobile functionality may interact with:

- Customer authentication data
- Personal information
- Authentication/session tokens
- Application programming interfaces (APIs)
- Transaction-related information
- Locally stored application data

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4. Recent Technology Changes

During the last 18 months, TechBridge Consulting Limited has undertaken several projects to support business growth:

- Migration of selected services to cloud infrastructure
- Development of new customer APIs
- Expansion of remote-working capabilities
- Introduction of additional mobile functionality
- Integration with additional payment providers
- Deployment of new web services
- Creation and modification of development and staging environments

Security resources have not increased at the same rate as technology development. Several teams have raised concerns that security assessments are sometimes performed late in the development lifecycle. The Head of Information Security has therefore requested a more structured and repeatable vulnerability-management approach.

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5. Existing Vulnerability Management Environment

TechBridge Consulting Limited does not currently operate a mature centralised vulnerability-management programme. Security activities are performed by different technology teams with varying levels of consistency.

Infrastructure administrators perform occasional vulnerability scans, developers conduct selected application testing, and external penetration tests are commissioned for some major projects. However, the organisation does not consistently follow a single lifecycle for vulnerability identification, validation, prioritisation, remediation, verification and closure.

A recent Internal Audit review identified the following programme-level concerns:

- No central vulnerability register
- Inconsistent vulnerability scanning
- Limited tracking of remediation
- Inconsistent severity classification
- No formally established remediation SLA
- No formal vulnerability exception process
- Limited evidence of remediation verification
- Vulnerability reports stored across different locations
- Insufficient management reporting
- Limited correlation of findings produced by different assessment tools

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6. Trigger for the Assessment

In August 2026, TechBridge Consulting Limited's Infrastructure monitoring platform detected unusual traffic originating from a server within its staging environment. The activity occurred outside normal working hours. The infrastructure team investigated but could not conclusively establish whether the traffic resulted from authorised testing, application behaviour or attempted exploitation.

No confirmed customer data breach was identified. However, the review established that the affected environment had not undergone a comprehensive vulnerability assessment for approximately nine months and that several components had been deployed or modified since the previous assessment.

At approximately the same time, the Mobile Development Team completed two Android application builds being considered for future production use. Senior management therefore commissioned a broader vulnerability assessment before approving the planned digital release.

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7. Your Engagement

You are part of the cybersecurity consulting team engaged by TechBridge Consulting Limited and have been assigned the role of Vulnerability Management Analyst.

Management Requirement

TechBridge Consulting Limited does not want separate scanner reports. Management requires a consolidated, evidence-based view of vulnerability exposure that explains what matters, why it matters, what should be remediated first and whether the planned production release should proceed.

Your assessment should enable management to answer:

- What vulnerabilities exist within the authorised assessment scope?
- Which findings represent genuine security weaknesses?
- Which vulnerabilities create the greatest business risk?
- What should be remediated first?
- Who should own the remediation actions?
- What longer-term vulnerability-management controls should be strengthened?
- Should the planned production release proceed?

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8. Authorised Assessment Targets

The instructor will provide the technical target information and assessment artifacts separately. For the engagement, treat the supplied systems and files as TechBridge Consulting Limited pre-production assets.

The resources to be assessed are attached as hyperlinks in Targets A, B and C. You will be redirected to a public OneDrive repo to download individual resources.

Target	Business Representation	Assessment Objective
Target A	TechBridge Consulting Limited Legacy Web Services Environment	Identify exposed services, web attack surface and security weaknesses requiring remediation.
Target B	TechBridge Consulting Limited Customer Mobile Application - Pre-Production Build	Assess the Android application for security weaknesses that could create unacceptable customer or business risk.
Target C	TechBridge Consulting Limited Digital Banking Application - Beta Build	Assess the Android application and determine whether identified security weaknesses should affect progression toward production.

Risk-Based Prioritisation Requirement

Scanner severity and CVSS are important inputs, but they must not be the sole basis for remediation priority. Consider business criticality, exposure, exploitability, data sensitivity, business function, compensating controls, regulatory consequences and potential operational impact.

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9. Asset Criticality and Data Classification

Asset / Business Service	Business Criticality	Data Sensitivity
Customer-facing web services	Critical	Confidential
Customer mobile services	Critical	Confidential
Digital banking application	High	Confidential
Customer authentication services	Critical	Restricted
Customer financial information	Critical	Restricted
Staging infrastructure	Medium	Internal
Development systems	Medium	Internal

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10. Approved Assessment Tools

You are not required to use every tool against every target. Tool selection must be technically appropriate to the asset being assessed. Supporting operating-system utilities may be used where reasonably necessary for analysis and evidence collection.

Tool	Approved Purpose
Nmap	Host discovery, port identification, service enumeration, version detection and attack-surface mapping.
Tenable Nessus	Vulnerability scanning, CVE identification, service vulnerability assessment, configuration observations and severity assessment.
Nikto	Web-server assessment, configuration weaknesses, exposed/default files and HTTP security observations.
OWASP ZAP	Web application discovery, passive analysis, authorised active scanning, application-security assessment and validation.
MobSF	Android APK static analysis, manifest and permission review, certificate analysis, code-security observations and insecure configuration review.

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11. Rules of Engagement

This is an authorised vulnerability assessment conducted in a controlled training environment. It is not an unrestricted penetration test.

Permitted Activities

- Network enumeration and service discovery
- Vulnerability scanning
- Web vulnerability scanning
- Mobile application static analysis
- Manual validation of selected findings
- Collection of evidence
- Limited interaction required to establish that a vulnerability exists

Out-of-Scope Activities

- Denial-of-service or resource-exhaustion testing
- Destructive exploitation
- Deletion or unauthorised alteration of data
- Persistence mechanisms or malware deployment
- Attacks against systems outside the supplied lab scope
- Scanning or testing public Internet targets
- Social engineering
- Credential attacks against external services

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Evidence Principle

A vulnerability does not need to be exploited to its maximum possible extent to be considered valid. Collect the minimum evidence reasonably required to demonstrate the issue while remaining within the Rules of Engagement.

12. Vulnerability Remediation Standard

For this engagement, TechBridge Consulting Limited has adopted the following interim remediation targets. These targets provide a baseline for planning; students must still apply risk-based judgement.

Severity	Standard Remediation Target
Critical	7 calendar days
High	30 calendar days
Medium	60 calendar days
Low	90 calendar days
Informational	Management discretion

When prioritising remediation, consider at minimum:

- Asset criticality
- External or internal exposure
- Exploitability
- Data sensitivity
- Business function
- Availability of public exploit information
- Existing or compensating controls
- Potential financial and operational impact
- Regulatory and contractual impact

13. Assignment Tasks

Task 1 - Attack Surface Discovery

Assess the supplied legacy web-services environment and establish its visible attack surface. Identify open ports, running services, service versions where reasonably determinable, exposed web services, potentially unnecessary services and observations requiring further investigation. Create an Asset and Service Inventory. An open port must not be classified as a vulnerability merely because it is open.

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Task 2 - Vulnerability Discovery

Perform vulnerability assessments using tools appropriate to each authorised target. Obtain and preserve relevant assessment evidence. Your work should demonstrate purposeful tool selection rather than indiscriminate scanning.

Task 3 - Finding Validation

Review scanner results before accepting them into the final vulnerability register. For material findings, determine whether sufficient evidence supports the issue. Distinguish confirmed findings, observations requiring further validation, false positives where identified, and informational results.

Task 4 - Consolidation and Deduplication

Combine findings from the assessment into one vulnerability register. Assign each accepted vulnerability a unique identifier. Do not create duplicate vulnerabilities simply because different tools report the same underlying weakness. Where appropriate, record multiple tools as evidence sources.

Task 5 - Vulnerability Risk Assessment

Assess each confirmed vulnerability in context. Consider the affected asset, vulnerability description, applicable CVE/CWE, evidence, CVSS, scanner severity, business impact, exploitability, asset criticality and final risk rating. Material changes to scanner-provided severity must be justified.

Task 6 - Risk-Based Prioritisation

Identify the Top 10 vulnerabilities requiring management attention. Do not simply select the ten highest CVSS scores. Explain why each vulnerability is prioritised using technical and business context.

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Task 7 - Remediation Planning

For each confirmed vulnerability, recommend a specific remediation action, responsible owner/team, remediation priority, applicable SLA/target date, compensating control where immediate remediation is not possible, and a method for verifying closure.

Task 8 - Vulnerability Management Improvement

Based on the engagement and TechBridge Consulting Limited's programme-level weaknesses, recommend practical improvements to the organisation's vulnerability-management lifecycle, including governance, scanning cadence, ownership, remediation tracking, exception handling, verification and management reporting.

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14. Required Vulnerability Register

Your consolidated vulnerability register must contain, at minimum, the following fields:

Required Field	Required Field
Vulnerability ID	Vulnerability Title
Affected Asset	Asset Criticality
Source Tool(s)	CVE/CWE (where applicable)
Vulnerability Description	Evidence
CVSS	Scanner Severity
Business Impact	Final Risk Rating
Remediation Priority	Recommended Remediation
Responsible Owner	SLA
Target Date	Status

Additional fields may be included where they improve the quality, traceability or usefulness of the assessment.

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15. Management Questions

1. What is TechBridge Consulting Limited's overall vulnerability posture?
2. What are the ten most important vulnerabilities identified?
3. Which asset or service presents the greatest current exposure, and why?
4. Were any vulnerabilities identified by multiple tools? Explain how they were consolidated.
5. Were any scanner findings rejected as false positives, treated as informational, or assigned a different priority? Explain your reasoning.
6. Which vulnerabilities could realistically create the greatest business impact?
7. Which findings require immediate remediation?
8. What vulnerability-management process weaknesses contribute to TechBridge Consulting Limited's current exposure?
9. What controls should TechBridge Consulting Limited introduce to establish a sustainable vulnerability-management programme?

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16. Production Release Recommendation

After completing the assessment, provide a formal recommendation to management concerning the planned production release. Select one of the following positions and justify it using evidence from your assessment:

Decision	Meaning
APPROVE	No material vulnerability identified within the assessed scope prevents release.
CONDITIONAL APPROVAL	Release may proceed subject to clearly defined remediation requirements and formal acceptance of residual risk.
DO NOT APPROVE	Current vulnerability exposure presents unacceptable risk and specified issues should be remediated before release.

Decision Requirement

There is no credit for selecting a release decision without justification. Your recommendation must be traceable to the vulnerabilities, business context, remediation status and residual risk identified during the engagement.

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17. Required Deliverables

Deliverable	Submission Requirement
1. Professional Vulnerability Assessment Report	A client-ready report containing executive summary, scope, methodology, attack-surface summary, risk overview, detailed validated findings, remediation roadmap, conclusion and release recommendation.
2. Vulnerability Management Register	An Excel workbook containing all validated and consolidated vulnerabilities, priorities, ownership, SLA/target dates, remediation actions, status and verification requirements.
3. Technical Evidence Pack	An organised collection of relevant screenshots and tool outputs supporting reported findings. Evidence must be labelled and traceable to the corresponding vulnerability ID.
4. Executive Security Brief	A maximum two-page management brief summarising overall vulnerability posture, Top 10 vulnerabilities, immediate actions and the production-release recommendation.

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18. Report Quality Expectations

Your submission should demonstrate the standard expected of a junior security consultant or vulnerability-management analyst. In particular:

- Do not submit raw scanner exports as the professional vulnerability assessment report.
- Do not report every scanner alert without review.
- Do not inflate the number of vulnerabilities by duplicating the same underlying weakness.
- Do not rely solely on scanner severity when establishing remediation priority.
- Clearly separate facts, evidence, professional judgement and assumptions.
- Use concise, technically accurate vulnerability titles and descriptions.
- Explain business impact in language a non-technical manager can understand.
- Make remediation recommendations specific and actionable.
- Ensure screenshots and evidence are readable, relevant and correctly referenced.
- Maintain consistent severity terminology and professional formatting throughout the submission.

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19. Suggested Assessment Weighting

Assessment Area	Weight
Discovery & Technical Assessment	20%
Validation & Evidence Quality	15%
Vulnerability Register & Deduplication	15%
Risk Assessment & Prioritisation	15%
Remediation Recommendations	15%
Professional Vulnerability Assessment Report	15%
Executive Communication & Release Decision	5%
Total	100%

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20. Submission Checklist

- ☐ All testing remained within the authorised scope.
- ☐ Asset and service inventory completed.
- ☐ Material scanner findings reviewed and validated.
- ☐ Duplicate findings consolidated.
- ☐ Vulnerability register completed.
- ☐ Top 10 vulnerabilities identified and justified.
- ☐ Remediation actions, owners and SLAs assigned.
- ☐ Evidence pack mapped to vulnerability IDs.
- ☐ Professional assessment report completed.
- ☐ Executive brief completed.
- ☐ Production-release recommendation clearly stated and justified.
- ☐ Files reviewed for professional presentation and consistency.

21. Professional Conduct Notice

The systems and artifacts supplied for this assignment are provided solely for authorised educational security testing. Techniques used during this capstone must not be directed at systems, applications or organisations for which you do not have explicit authorisation.

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